Afsar Tanvir

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Education

B.Sc. in Computer Science and Engineering Jan 2020 – Jul 2025
Sylhet Engineering College *Sylhet, Bangladesh*
Affiliated university: Shahjalal University of Science and Technology (SUST), Sylhet
CGPA: 3.68 / 4.00

Academic Interests

My primary interest is in distributed, cloud and data-intensive systems, in particular scheduling, caching, consistency and databases. It grew out of production work on a multi-tenant community platform, where tenant isolation, cache invalidation and pagination are everyday correctness concerns, and out of building a self-hosted job scheduler and an API mock server backed by a Redis cache. A secondary interest is software engineering and developer tooling, especially API mocking and testing and the design of testable systems; earlier, I explored natural language processing by building an LSTM language model. I am seeking an MSc, either taught with a substantial dissertation project or research-based, to study these areas in depth.

Professional Experience

Full Stack Developer Sep 2025 – Present
AppifyLab *Bangladesh*

- Develop EzyCommunity, a multi-agency, multi-tenant community platform on which agencies run many independent communities: AdonisJS (Node.js/TypeScript) API and Next.js/React frontend over MySQL, MongoDB and Redis; 550+ non-merge commits since Feb 2026.
- Implemented a content-moderation workflow: a new moderator permission in the role-based access-control (RBAC) system, soft deletes, review queues for pending posts and pending edits, and community-scoped queries that enforce tenant isolation.
- Improved the real-time home and channel feeds: a "new posts" pill that never inserts live posts unasked, infinite-scroll performance and layout-shift fixes, and cursor pagination for comments.
- Improved Redis channel-cache invalidation; built a website-builder settings area with screens for fonts, colour variables and custom CSS/JS.
- Contributed to Stripe-backed checkout for paid channels; shipped invite links, gamification, an image cropper and notification unsubscribe.

.NET Developer Trainee Dec 2024 – Apr 2025
Learnathon by Geeky Solutions – Team "Brainstormers" *Certificate*

- Built TaskForge.NET, a task-management system in ASP.NET Core MVC following Clean Architecture and SOLID principles, with task CRUD, user assignment and status tracking (To Do / In Progress / Done).
- Persisted data in SQL Server through Entity Framework Core; worked in a team of three using a GitHub pull-request workflow with code review.

Selected Projects

MockFlow | *AdonisJS 6 (TypeScript), Next.js, PostgreSQL, Redis* May 2026 – Present

- Collaborative API mock-server platform that lets frontend and QA teams design, document and serve mock REST endpoints before the backend exists.
- Mock engine: path-to-regexp route matching ordered by route specificity, faker.js response templates, and scenarios and rules with randomised delay.
- OpenAPI 3.x and Postman v2.1 export plus a two-phase (preview → apply) import; version-history snapshots with an automatic backup before every restore; request logs with keyset pagination.
- Companies and teams with role-based access, and a super-admin console with impersonation.
- Redis caching with single-flight, negative caching, TTL jitter and tag-based invalidation; strict Controller → Service → Query → Model layering.

- ∼178 unit and ∼75 integration tests (Japa) running against an isolated test database.
- Security hardening: rate-limited public auth routes, a 1 MB body cap, a race-safe last-admin check and no account enumeration on password reset.

local-scheduler | *Go, SQLite, Docker* Oct 2026 – Present

- Self-hosted job scheduler: one binary with embedded SQLite and an HTTP/JSON API (OpenAPI 3); cron or one-time jobs with time zones run HTTP calls or shell commands.
- Execution semantics: a no-overlap rule, a single catch-up run for missed schedules, graceful shutdown that drains running jobs for 30 s, and crash recovery that marks interrupted runs as failed.
- Unit tests plus an end-to-end script that drives the real binary through restarts, `kill -9` recovery, timeouts and memory checks.

Audio Visualizer | *AdonisJS 7, Next.js 16, React 19, MongoDB, Web Audio API* Mar 2026 – May 2026

- Real-time frequency-bar and waveform rendering for MP3, WAV and FLAC files with the Web Audio API (live demo).
- Upload, streaming and delete endpoints; drag-and-drop upload, playback controls and upload history.

Next Word Predictor | *Python, TensorFlow/Keras, Flask* Jun 2025

- LSTM language model for next-word prediction: tokenisation, padding and n-gram sequence generation for training; evaluated with top-1 and top-5 accuracy.
- Temperature sampling for more varied suggestions, served through a small Flask web interface.

Competitive Programming & Awards

- 1800+ problems solved across online judges.
- Codeforces: Specialist, maximum rating 1476 | CodeChef: 3-star, rating 1744.
- **14th place**, MU Inter-University Programming Contest (IUPC) 2024, Sylhet Division.
- **89th place**, DUET IUPC 2025.
- **97th place**, National Collegiate Programming Contest (NCPC) 2023, onsite at Jahangirnagar University (held March 2024).

Leadership & Service

Coordinator, SEC Programming Club 2024 – 2025
Sylhet Engineering College *Sylhet, Bangladesh*

- Organised internal selection contests and warm-up sessions; managed team registration for inter-university contests.

Technical Skills

Languages: TypeScript, JavaScript, Go, C#, C++, Python, SQL
Backend: AdonisJS, Node.js, ASP.NET Core, REST API design, OpenAPI
Frontend: Next.js, React, Tailwind CSS
Data: PostgreSQL, MySQL, MongoDB, Redis, SQLite
Practices: Automated testing (Japa, Go test), Clean Architecture, SOLID
Tools: Git/GitHub, Docker, Linux

Languages

Bengali: Native
English: Professional working proficiency

References

Available upon request.